UNIVERSITÀ DI TORINO

Analisi e Visualizzazione delle Reti Complesse

NS06 - Centrality Measures

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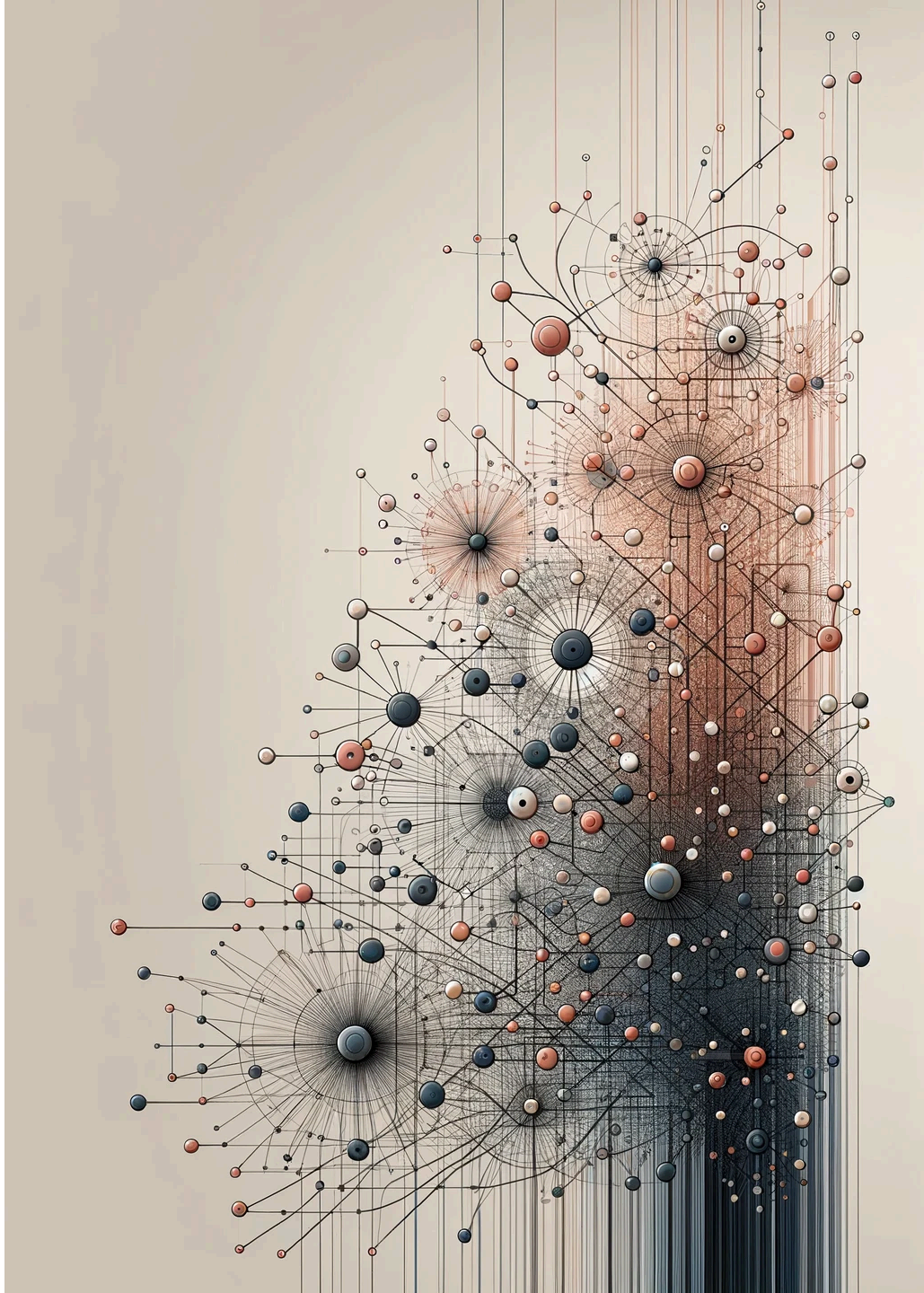
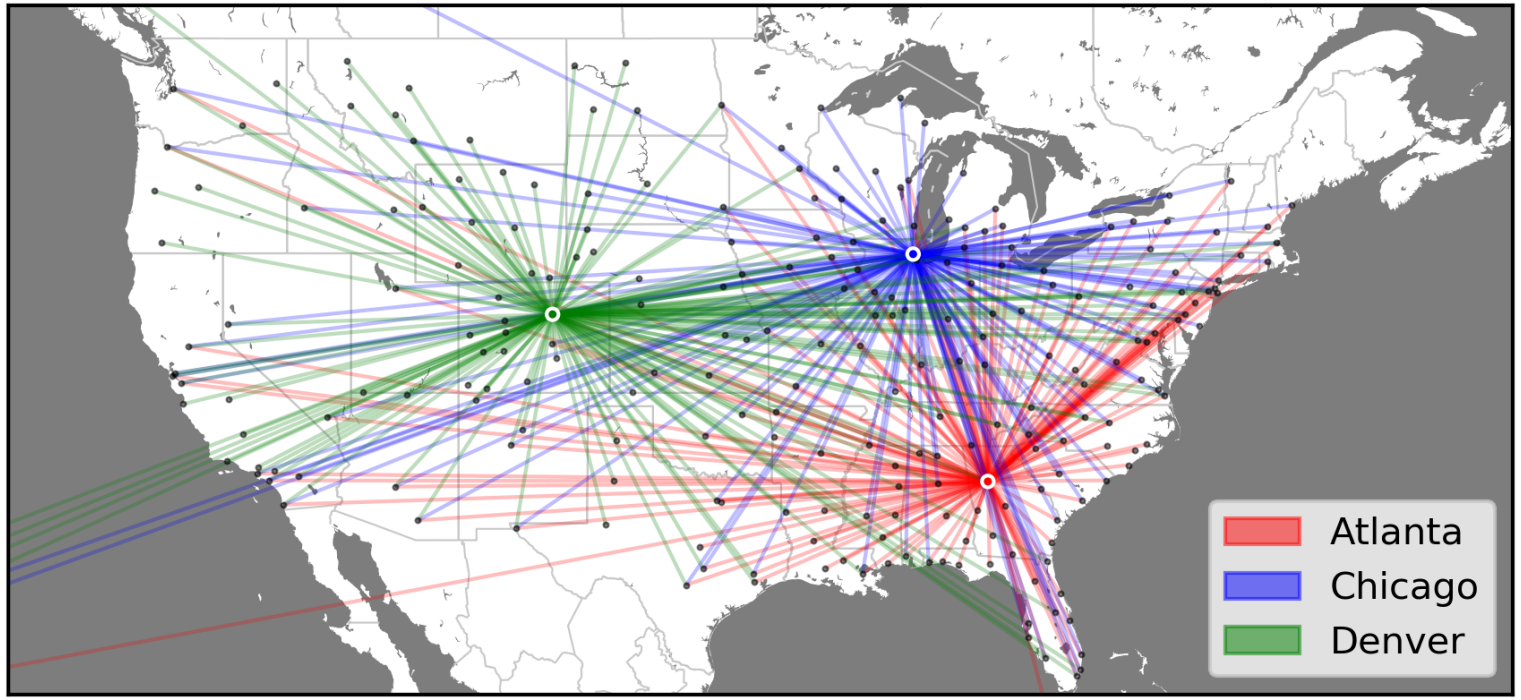


Outline

- **Degree centrality** — local connectivity; hubs
- **Eigenvector centrality** — recursive importance; quality of connections
- **Katz centrality** — walk-counting with baseline; directed networks
- **PageRank & HITS** — preview only; full treatment in `NS19`
- **Closeness centrality** — proximity to all nodes; information diffusion
- **Harmonic centrality** — closeness for disconnected graphs
- **Betweenness centrality** — control over shortest paths; bridges
- **k-core decomposition** — structural embeddedness; cohesive core

Real networks are heterogeneous



Some nodes (and links) are much more important (**central**) than others!

Centrality measures

- **Centrality**: it measures the **importance** of a node in a network
- It can be quantified by different **measures**:
  - **Degree**: The number of direct connections a node has. High-degree nodes are often referred to as **hubs**.
  - **Eigenvector**: Considers not just the quantity but the quality of connections. A node is important if it is connected to other important nodes.
  - **Katz**: An extension of eigenvector centrality that counts walks of all lengths between nodes, with longer walks given lower weight.
  - **Closeness**: Measures how close a node is to all other nodes in the network. A node with high closeness centrality can quickly interact with all other nodes.
  - **Betweenness**: Quantifies the number of times a node acts as a bridge along the shortest path between two other nodes. Nodes with high betweenness centrality have influence over the flow of information in the network.

Degree

- The **degree of a node** k_i is the **number of neighbors** of the node i
- High-degree nodes are called **hubs**
- **Average degree** of an undirected network:

$$\langle k \rangle = \frac{\sum_i k_i}{N} = \frac{2L}{N}$$

In NetworkX:

```
G.degree(2) # returns the degree of node 2

G.degree()   # returns a dict with the degree of all nodes of G
```

Reference: Freeman, L.C. (1979). *Centrality in social networks: Conceptual clarification*. Social Networks, 1(3), 215–239.

Applications

Social Networks

- **Influencers Identification**: In social media platforms like Twitter or Facebook, users with a high degree centrality are often influencers who can spread information quickly.

Biology

- **Protein-Protein Interaction Networks**: In biological networks, proteins with high degree centrality are often essential for the survival of an organism.

Transportation

- **Airport Networks**: Airports with high degree centrality are major hubs that connect many flights, playing a crucial role in the efficiency of air travel.

However, degree centrality is a purely local measure that only considers direct connections, potentially overlooking nodes that serve as critical bridges between different parts of the network despite having fewer connections.

The limitation of degree centrality

Consider two nodes, both with degree $k = 3$:

- **Node A**: connected to three high-degree hubs (each with degree 100)
- **Node B**: connected to three leaves (each with degree 1)

Degree centrality assigns identical scores to A and B. Yet A sits at the centre of the most connected region of the network, while B is peripheral.

The limitation is structural: degree counts neighbours but ignores *who those neighbours are*.

This is the precise gap that eigenvector centrality addresses: a node's importance should be a function of the importance of its neighbours — a recursive definition that propagates through the entire network.

Eigenvector

- With degree centrality, we treat all neighbors equally regardless of their importance
- In many contexts, **a node is more important if it connects to other important nodes**
- Eigenvector centrality captures this recursive notion of importance

For a node i, its eigenvector centrality x_i is defined as:

$$x_i = \frac{1}{\kappa} \sum_{j=1}^{n} A_{ij} x_j$$

Where:

- x_j is the centrality of neighboring node j
- A_{ij} is the adjacency matrix entry (1 if connected, 0 if not)
- κ is a proportionality constant

Matrix Formulation

The equation can be written in matrix form:

$$\mathbf{x} = \frac{1}{\kappa} A\mathbf{x}$$

Which rearranges to the standard eigenvalue equation:

$$A\mathbf{x} = \kappa\mathbf{x}$$

$\mathbf{x}$ is an eigenvector of the adjacency matrix A

Moreover:

1. since A is a non-negative matrix
2. we want $\mathbf{x}$ to be non-negative

because of the **Perron-Frobenius theorem** $\mathbf{x}$ is the **leading eigenvector**

(Perron-Frobenius theorem states that for a matrix with all elements non-negative, like the adjacency matrix, there is only one eigenvector that also has all elements non-negative, and that is the leading eigenvector)

Reference: Bonacich, P. (1972). *Factoring and weighting approaches to status scores and clique identification*. Journal of Mathematical Sociology, 2(1), 113–120.

- **Then the eigenvector centrality x_i node i is the ith element of the leading eigenvector of the adjacency matrix**
- **κ must be equal to the largest eigenvalue**
- Normalization could be applied, e.g., centralities sum to n (which ensures that average centrality stays constant as the network gets larger)
- Technically, this result is only true for connected networks, i.e., networks with only one component.
    - If a network has more than one component, then there is one eigenvector with nonnegative elements for each component.

```
nx.eigenvector_centrality(G)   # returns the eigenvector centrality for the graph G
```

- This works for only for undirected network

Why the Largest Eigenvalue?

- For eigenvector centrality, we use the eigenvector corresponding to the largest eigenvalue (principal eigenvector) because:
  - It represents the most stable solution to the recursive definition of importance
  - It's the only eigenvector guaranteed to have all non-negative values (by the Perron-Frobenius theorem)
  - It captures the most dominant pattern of connectivity in the network
- **Simple explanation:** The principal eigenvector identifies nodes that are:
  i. Connected to many other nodes, or
  ii. Connected to a few very important nodes, or
  iii. Some combination of both
- This provides a natural way to identify influential nodes based not just on how many connections they have, but on the quality of those connections.

Eigenvector centrality: derivation on a path graph

Path P_4: nodes $\{1, 2, 3, 4\}$, edges $\{(1, 2), (2, 3), (3, 4)\}$.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

The leading eigenvector is proportional to $\left[\sin\frac{\pi}{5},\ \sin\frac{2\pi}{5},\ \sin\frac{3\pi}{5},\ \sin\frac{4\pi}{5}\right] \approx [0.588,\ 0.951,\ 0.951,\ 0.588]$.

Ranking: $2 \approx 3\ >\ 1 \approx 4$.

Degree gives $k_1 = k_4 = 1$, $k_2 = k_3 = 2$ — the same qualitative order. But eigenvector centrality encodes the difference quantitatively: the ratio of centrality scores ($\approx 1{:}1.618$, the golden ratio) reflects the spectral structure of the path, not just the count of connections.

Why Eigenvector Centrality Fails for Directed Networks

Problem: Source Nodes

- In our example network:

```
A → B → C
        ↓
D ← E ← F
```

- Recall the eigenvector centrality equation: $x_i = \frac{1}{\lambda} \sum_j A_{ji} x_j$
  - $A_{ji} = 1$ if there's an edge from j to i
  - $A_{ji} = 0$ otherwise
- For node A:
  - No incoming edges means all $A_{ji} = 0$
  - Therefore $x_A = 0$ (zero centrality)
- This zero propagates: if $x_A = 0$, then x_B becomes zero, then x_C, etc.

Katz Centrality

- **Extends eigenvector centrality** to solve problems with directed networks
- **Problem with eigenvector centrality in directed networks:**
  - Nodes with no incoming edges receive zero centrality
  - This creates a "zero centrality problem" that cascades through the network
- **Katz's solution:** Add a baseline importance to every node

$$x_i = \alpha \sum_{j=1}^{n} A_{ij} x_j + \beta$$

Where:

- x_i is the centrality of node i
- α is an attenuation factor (smaller than the reciprocal of largest eigenvalue)
- β is the baseline centrality given to each node (typically 1)
- A_{ij} is the adjacency matrix element (1 if there's an edge from j to i)
- In matrix form: $\mathbf{x} = \alpha A \mathbf{x} + \beta \mathbf{1}$
- Solution: $\mathbf{x} = \beta (I - \alpha A)^{-1} \mathbf{1}$

Katz Centrality

- **Walks-based interpretation**: Katz centrality can be interpreted as **counting walks of all lengths** from a node, but with longer walks attenuated by powers of α
- **Mathematical interpretation**: When we solve $\mathbf{x} = \alpha A\mathbf{x} + \beta\mathbf{1}$ to get $\mathbf{x} = \beta(I - \alpha A)^{-1}\mathbf{1}$, this expands to:

$$\mathbf{x} = \beta(I + \alpha A + \alpha^2 A^2 + \alpha^3 A^3 + ...)$$

  Where:

  - A^k counts walks of length k between nodes
  - α^k reduces the importance of longer walks

- **Parameter significance**:
  - $\alpha < \frac{1}{\lambda_{max}}$ ensures convergence (where λ_{max} is the largest eigenvalue of A)
  - Smaller α values prioritize shorter walks
  - Larger α values (while still below the threshold) give more weight to longer walks
- **Relationship to other measures**:
  - When $\alpha \to 0$: Approaches degree centrality (only direct connections matter)
  - When α approaches $\frac{1}{\lambda_{max}}$: Approaches eigenvector centrality

In networkX:

```
nx.katz_centrality(G, alpha=0.1, beta=1.0)
```

Katz centrality: social status and prestige

Katz (1953) introduced the measure in the context of *social status* within a group: an individual's prestige derives not simply from how many others endorse them, but from the prestige of those who do.

Citation network analogy: a paper cited by highly cited papers accrues more Katz centrality than one with the same number of citations from obscure sources.

The parameter β resolves a structural pathology of eigenvector centrality: in a directed network, nodes with no incoming edges receive zero eigenvector centrality — and this zero propagates to all their successors. The baseline β breaks the cascade: every node carries a non-zero initial status that cannot be annihilated by the absence of incoming links.

This makes Katz centrality the first measure in this lineage that is well-defined on directed networks with source nodes.

Katz vs. Eigenvector Centrality: When to Use

Use Katz centrality when:

1. You're working with **directed networks**
2. Your network contains **source nodes** (with no incoming edges)
3. Your network has **multiple components**
4. You want to **control the extent of influence propagation** using α

```
# Computing Katz centrality for a directed graph
G = nx.DiGraph([(0,1), (1,2), (2,3), (3,0)])
centrality = nx.katz_centrality(G, alpha=0.1)
```

Preview: from Katz to PageRank and HITS

Directed centralities follow a simple progression:

- **Eigenvector**: importance flows from important predecessors, but source nodes may collapse to zero
- **Katz**: adds a baseline β, so every node retains a non-zero score
- **PageRank** and **HITS**: directed refinements for information networks

Preview: PageRank and HITS

Two influential refinements:

- **PageRank**: normalizes endorsements by source out-degree; damping guarantees convergence
- **HITS (hub)**: rewards nodes that point to good authorities
- **HITS (authority)**: rewards nodes pointed to by good hubs

```
nx.pagerank(G, alpha=0.85)   # Brin & Page, 1998
nx.hits(G)                   # Kleinberg, 1999 — returns (hub_dict, authority_dict)
```

Treat these as a preview. Full treatment in **NS19 - Link Analysis and Web Search**.

Closeness

Idea: **A node is more central the closer it is to the other nodes, on average**

$$g_i = \frac{1}{\sum_{j \neq i} l_{ij}}$$

where l_{ij} is the distance between nodes i and j

In a normalized form (discounting the graph size):

$$g_i = \frac{N-1}{\sum_{j \neq i} l_{ij}} = \frac{1}{\frac{\sum_{j \neq i} l_{ij}}{N-1}}$$

That is the inverse of the average distance from the focal node i to the rest of the network

```
nx.closeness_centrality(G, node)   # returns the closeness centrality of node i
```

Closeness: worked example

Compute closeness directly from shortest-path distances.

Node 1 (geometric center):

$$\sum_j l_{1j} = 1 + 1 + 1 + 1 + 2 = 6 \quad \Rightarrow \quad g_1 = \frac{5}{6} \approx 0.83$$

Smallest total distance $\Rightarrow$ highest closeness.

Closeness: center vs. periphery

For node 2:

$\sum_j l_{2j} = 1 + 2 + 2 + 2 + 3 = 10 \quad \Rightarrow \quad g_2 = \frac{5}{10} = 0.50$

For node 6:

$\sum_j l_{6j} = 2 + 3 + 3 + 3 + 1 = 12 \quad \Rightarrow \quad g_6 = \frac{5}{12} \approx 0.42$

Ranking: $g_1 > g_2 = g_3 = g_4 = g_5 > g_6$.

Moving away from the center increases the total distance and lowers closeness.

Closeness is geometric: a node may have moderate degree yet be central if it minimizes average distance.

Closeness Centrality: Key Applications

Public Health

- **Disease Control**: Optimal locations for vaccination centers to minimize epidemic response time
- **Healthcare Facility Placement**: Positioning medical facilities to reduce average patient travel time

Infrastructure & Emergency Services

- **Facility Location**: Optimal positioning for fire stations, distribution centers
- **Network Design**: Identifying critical infrastructure expansion points

Social Networks

- **Information Diffusion**: Finding individuals who can quickly spread information
- **Influence Maximization**: Identifying efficient "influencers" with broad reach

When Most Valuable

- When **speed of dissemination** is critical
- When **minimizing time delays or costs** across a system
- In **connected networks** where all nodes can reach each other

Limitations

- Less meaningful in **disconnected networks** (infinite distances between components)
- Sensitive to **network boundaries** and size
- Computationally expensive for very large networks

Harmonic centrality

Closeness centrality is undefined on disconnected graphs: if node j is unreachable from i, then $l_{ij} = \infty$ and $g_i = 0$ regardless of the node's connectivity to its reachable component.

Marchiori & Latora (2000) proposed replacing the sum of distances with the sum of their reciprocals:

$$H_i = \sum_{j \neq i} \frac{1}{l_{ij}}$$

By convention $1/\infty = 0$, so unreachable pairs contribute nothing. On a connected graph H_i is monotonically related to g_i; on a disconnected graph H_i remains well-defined and structurally meaningful.

Difference from closeness: identical intuition (central nodes are close to everyone) but applicable to any graph topology. No special-casing for disconnected components is required.

Boldi & Vigna (2014) established the axiomatic foundations. `nx.harmonic_centrality(G)`.

Betweenness

Idea: **A node is more central the more often it is crossed by shortest paths**

$$
\begin{aligned}
b_i &= \sum_{h \neq j \neq i} \frac{\sigma_{hj}(i)}{\sigma_{hj}} \\
\sigma_{hj} &= \text{number of shortest paths from } h \text{ to } j \\
\sigma_{hj}(i) &= \text{number of shortest paths from } h \text{ to } j \text{ running through } i
\end{aligned}
$$

This measure **depends on the size of the network**.

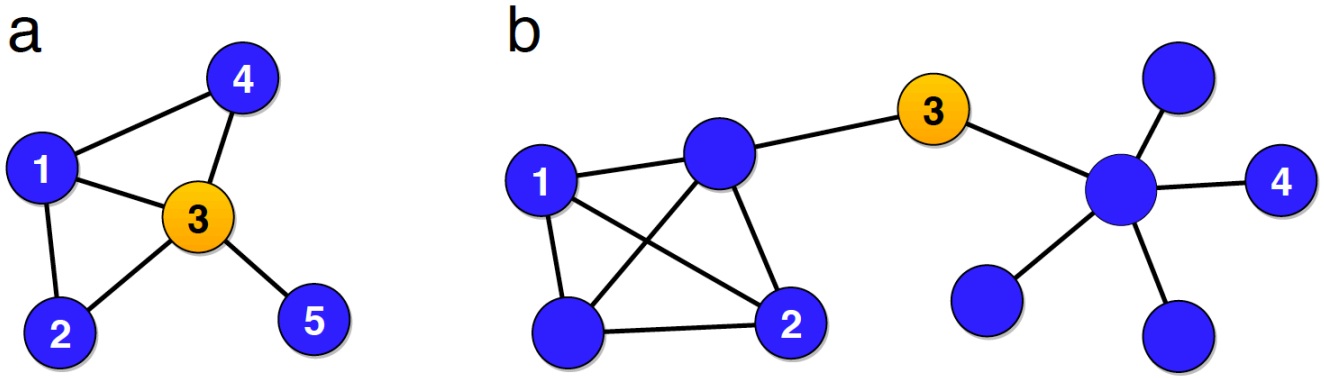
We can normalize the formula by the **maximum number of paths that could go through node** i, e.g., the number of pairs of nodes excluding i:

$$
\frac{b_i}{\binom{N-1}{2}} = \sum_{h \neq j \neq i} \frac{2\sigma_{hj}(i)}{(N-1)(N-2)\sigma_{hj}}
$$

Betweenness

Hubs usually have a **high betweenness** (a).

There can be **nodes with high betweenness that are not hubs** (b).



Edge Betweenness

Betweenness can be easily extended to edges.

Idea: **The fraction of shortest paths among all possible node pairs that pass through the link**

```
nx.betweenness_centrality(G)       # dict nodes: betweenness centrality

nx.edge_betweenness_centrality(G)  # dict links: betweenness centrality
```

Reference: Brandes, U. (2001). *A faster algorithm for betweenness centrality*. Journal of Mathematical Sociology, 25(2), 163-177. *(The $O(VE)$ algorithm implemented by NetworkX.)*

Edge betweenness and communities

Edges between communities lie on many inter-community shortest paths.

They therefore accumulate high **edge betweenness**.

Boundary edges are structural bottlenecks; edge betweenness measures exactly this role.

Girvan-Newman idea

Use edge betweenness as a divisive rule:

1. compute edge betweenness
2. remove the highest-scoring edge
3. recompute betweenness
4. repeat

Communities emerge as bottleneck edges are removed.

Full treatment in **NS13 - Community Detection**.

Reference: Girvan, M. & Newman, M.E.J. (2002). *Community structure in social and biological networks*. PNAS, 99(12), 7821–7826.

Beyond paths

Structural embeddedness in dense cores

k-core and coreness

Centrality can also mean **structural embeddedness**, not only reach or brokerage.

The k-core asks which nodes remain inside a cohesive subgraph after repeated pruning.

The **k-core** is the maximal induced subgraph in which every node has degree $\geq k$.

The **coreness** of node i is:

$$c_i = \max\{k : i \in \text{k-core}(G)\}$$

Computing the k-core

Iteratively remove nodes whose current degree is below k.

This peels the graph from sparse periphery to dense interior.

```
nx.core_number(G)   # {node: coreness},  O(m) algorithm
nx.k_core(G, k=3)   # induced subgraph of the 3-core
```

Shell interpretation:

- low coreness: peripheral nodes
- high coreness: deeply embedded nodes

Coreness in relation to degree and betweenness

Coreness vs. degree: a star centre has degree $N-1$ but coreness 1 (all its neighbours are leaves, forming no k-core with $k \geq 2$). A node inside a dense clique has moderate degree but maximum coreness.

Coreness vs. betweenness: a bridge between two communities has high betweenness (all inter-community paths cross it) but low coreness (it lies at the periphery of both communities). Core nodes score high in coreness but may not lie on many shortest paths.

Why it matters:

- **Epidemic spreading** — Kitsak et al. (2010) demonstrated across multiple empirical networks that coreness predicts spreading influence more accurately than degree or betweenness: the most effective spreaders are those in the innermost core, not necessarily the highest-degree nodes
- **Network robustness** — the k-core is the last subgraph to fragment under targeted attacks (NS07)
- **Epidemic thresholds** — core structure governs the onset of large outbreaks (NS22–23)

Centrality measures at a glance

Measure	What it measures	Useful when	What it ignores	Typical failure mode
Degree	local connectivity	first-pass hubs, local popularity	neighbor quality, global position	over-rates star centres, misses bridges
Eigenvector	recursive importance	prestige in mutually reinforcing neighborhoods	brokerage, disconnectedness	collapses on directed graphs with source nodes
Katz	attenuated walks plus baseline	directed networks and prestige with source nodes	geometric position, pure brokerage	result depends on the choice of α
Closeness	inverse average distance	fast access or broadcast in connected graphs	bottlenecks, cohesion	misleading or undefined on disconnected graphs
Harmonic	reachability in disconnected graphs	partially reachable or disconnected graphs	cohesion, shortest-path control	does not identify brokers

Centrality measures at a glance

Measure	What it measures	Useful when	What it ignores	Typical failure mode
Betweenness	shortest-path brokerage	bottlenecks, mediators, bridges	dense-core embeddedness, non-shortest routes	unstable when many alternative paths exist
Coreness	structural embeddedness	dense cores, spreaders, resilient inner structure	geometry, brokerage	misses bridge nodes and star centres

No single measure is best: the right choice depends on the structural role you care about.

`PageRank` and `HITS` adapt recursive centrality to directed information networks; they return in `NS19`.

Additional readings

- Linton C. Freeman, Centrality in social networks. A conceptual clarification, Social networks, vol. 1, no 3, 1979, p. 215–239
- Ulrik Brandes (2001) A faster algorithm for betweenness centrality, The Journal of Mathematical Sociology, 25:2, 163-177
- Girvan, M. & Newman, M.E.J. (2002). Community structure in social and biological networks. PNAS, 99(12), 7821–7826
- Kitsak, M. et al. (2010). Identification of influential spreaders in complex networks. Nature Physics, 6, 888–893
- Marchiori, M. & Latora, V. (2000). Harmony in the small-world. Physica A, 285(3–4), 539–546

Reading material

References

[ns1] **Chapter 3**

[ns2] **Easley & Kleinberg, Chapter 2**

Q&A